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Chapter-4

Errors:

(1) Inherent Error:-

Inherent Error is the error that already exists in the given data before any calculation is done. The computer doesnot create this error. The error comes from the input data itself.

Types of Errors:

(1) Data Error: Occurs during measurement or data collection.

(2) Converation Error: Occurs when numbers are converted (for example decimal to binary) and cannot be represented exactly.

Ex: If a weighing machine shows 50.2 kg, the real weight maybe 50.18 kg or 50.23 kg So the error already exists is inherent.

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Ex 4.3 Represent the decimal numbers 0.1 and 0.4 in binary form with an accuracy of 8 binary digits. Add them and then convert the result back to decimal form.

$\Rightarrow 0.1_{10}$

$\Rightarrow 0.1 \times 2 = 0.2 \quad 0$
$\rightarrow 0.2 \times 2 = 0.4 \quad 0$
$\rightarrow 0.4 \times 2 = 0.8 \quad 0$
$\rightarrow 0.8 \times 2 = 1.6 \quad 1$
$0.6 \times 2 = 1.2 \quad 1$
$0.2 \times 2 = 0.4 \quad 0$
$0.4 \times 2 = 0.8 \quad 0$
$0.8 \times 2 = 1.6 \quad 1$

$(0.1)_{10} = (0.00011001)_2$

0.4_{10}

$0.4 \times 2 = 0.8 \quad 0$
$0.8 \times 2 = 1.6 \quad 1$
$0.6 \times 2 = 1.2 \quad 1$
$0.2 \times 2 = 0.4 \quad 0$
$0.4 \times 2 = 0.8 \quad 0$
$0.8 \times 2 = 1.6 \quad 1$
$0.6 \times 2 = 1.2 \quad 1$
$0.2 \times 2 = 0.4 \quad 0$

$\therefore (0.4)_{10} = (0.01100110)$

$0.1_{10} = 0.00011001$
$0.4_{10} = 0.01100110$

0.01111111

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Date:...../...../.....

$(0.01111111) = 0 \times 2^{-1} + 1 \times 2^{-2} + 1 \times 2^{-3} + 1 \times 2^{-4} + 1 \times 2^{-5} + 1 \times 2^{-6} + 1 \times 2^{-7} + 1 \times 2^{-8}$

$= 0.49609375$

$\therefore (0.1 + 0.4) = 0.5$

Answers should be 0.5 but it is not.

This is due to the errors in conversion from decimal to binary.

Numerical Errors:

Computers cannot store exact values all the time, so small mistakes happen. These mistakes are called Numerical errors.

Two types:

1. Roundoff Errors
2. Truncation

Roundoff errors:

Roundoff errors happens when we keep only a limited number of digits and change the original number.

Ex :- $3.1416 \rightarrow 3.14$

Errors $= 3.1416 - 3.14$

$= 0.0016$

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Two type of roundoff:

chopping is the process is ?

① **Chopping**: Just remove extra digits (no rounding).

Example: Exact = 45.6789

Stored = 45.67

Error = 45.6789 − 45.67

= 0.0089.

~~② Truncation Errors~~

Error = $g_x \times 10^{E-d}$ where $0 \leq g_x < 1$

The maximum value of g_x is less than 1.0

true $x = (f_x + g_x \times 10^d) 10^F$

Absoulate error $\leq 10^{F-d}$

E = Exponent
d = length of mantissa
f_x = mantissa
g = dropped part

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Date:...../...../.....

Symmetric Round off: is a rounding where errors are equally likely to be positive or negative, reducing bias

true $x = f_x \times 10^E + g_x \times 10^{E-d}$

Error $= g_x \times 10^{E-d}$ $[g_x < 0.5)$

Approximate, $x = f_x \times 10^E$ $[g_x < 0.5]$

Note: $g < 5$ হলে No change

$g \geq 5$ " last digit এর আগের 1 টি

$42.7329 \rightarrow 42.73$

$42.7329 \rightarrow 42.733$

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$[fx \times 10^E + gx \times 10^{E-d}]$

Ex 4.4: find the roundoff error in ~~sorting~~ storing the number 752.6835 using a four digit Mantissa.

⇒ ~~True x = 0.7526 × 10^3 + 0.835 × 10^-1~~

752.6835

$= 0.\underline{7526}\underline{835} \times 10^3$

$= (0.7526 + 0.\underline{0000}\,835) \times 10^3$
$\quad\quad\quad

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Date:...../...../.....

Approximate, ~~Error~~ $x = 0.752 \times 10^3$

* Truncation Errors:

Truncation Error happens when we use an approximate Method instead of the exact mathematical process

Example: Series expansion of $\sin(x)$:

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

We cannot use all terms, so we stop after a few terms.

The ignored terms = truncation error.

Ex 1.5 find the truncation error in the result of the following function for $x = 1/5$ when we use a) first three terms b) first four terms and c) first five terms.

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \frac{x^6}{6!} -$$

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a) for first 3 terms,

Truncation error = e^x approx, 7 terms − e^x approx, 3 terms.

$= e^x$ approx, last 4 terms

for $x = \frac{1}{5} = 0.2$

$$\text{Truncation error} = \frac{(0.2)^3}{3!} + \frac{(0.2)^4}{4!} + \frac{(0.2)^5}{5!} + \frac{(0.2)^6}{6!}$$

$$= 1.402755 \times 10^{-3}$$

$$= 0.1402755 \times 10^{-2}$$

b) for first 4 terms,

T.E = e^x approx, 7 terms − e^x approx, 4 terms

$= e^x$ approx, last 3 terms.

$$\therefore \text{T.E} = \frac{(0.2)^4}{4!} + \frac{(0.2)^5}{5!} + \frac{(0.2)^6}{6!}$$

$$= 0.69422 \times 10^{-4}$$

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Date:...../ ...../ .....

c) for first 5 terms.

T.E = e^x approx, 7 terms − e^x approx, 5 terms

= e^x approx, last 2 terms.

for $x = \frac{1}{5} = 0.2$

$\therefore$ T.E = $\frac{(0.2)^5}{5!} + \frac{(0.2)^6}{6!}$

$= 0.2755 \times 10^{-5}$.

Ex; 4.6 for $x = -\frac{1}{5}$.

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